Venkatraman SK

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Education

PSG College of Technology Jun 2018 - May 2023
MSc Software Systems *Coimbatore, TamilNadu*

Experience

Lassie AI December 2025 – Present
Software Engineer – AI *San Francisco, CA (Remote)*

- **PMS Sync:** Built the **PMS sync ecosystem** to fetch and synchronize patient and insurance data **from on-premise Windows servers into Lassie's GCP infrastructure**, enabling reliable data availability for downstream AI and application workflows
- Introduced **fingerprint-based incremental synchronization** to efficiently detect updates and deletes from on-premise PMS servers, **reducing end-to-end synchronization time from 2 hours to 6–10 minutes**
- **AI Poster:** Built an **AI-powered insurance payment posting workflow** to automate the manual process of accounting for paid insurance claims in dental practices' PMS systems
- Improved the long tail of patient and claim **matching success rates from 91% to 98%** by enhancing matching workflows and handling variations in patient, insurance, and claim data across PMS systems
- Designed and implemented **write-tools for multiple PMS ecosystems**, including Dentrix, OpenDental, and Eaglesoft, enabling the AI posting workflow to perform system-specific actions across different practice management systems
- **Data Monitoring & Reporting:** Built and maintained materialized views on top of existing PMS data **to generate practice payer profiles and payer-status reporting**, providing business-level visibility into payer availability
- Developed **batch jobs** to generate and maintain practice-level payer profiles and canonical payer-level aggregations, enabling better decision-making around payer enrollments and AI posting workflow
- **Evals and Developer Tooling:** Built **evaluation workflows** to measure the quality and reliability of the AI-powered posting planner and **identified and developed a golden dataset** to provide a consistent benchmark for evaluating poster performance and prevent regressions
- **Infrastructure:** Built and maintained **asynchronous processing infrastructure using Redis-backed task queues** to support reliable execution of background processing workflows
- Optimized Cloud Run services through resource rightsizing and request-based provisioning, reducing infrastructure costs by **$2,000 per month** while maintaining application performance
- Mentored junior engineers, helping them navigate system design and technical trade-offs in building production systems

Tech Stack: Python, Django, Pydantic AI, PostgreSQL, Redis, GCP (Cloud Run, Cloud Tasks, Cloud Storage)

AppsForBharat May 2023 – November 2025
Software Engineer 2 – Backend and data platform *Bengaluru, India*

- **Backend & Data Platform:** Built and maintained Go microservices for **event capture and data ingestion**, enabling low-latency and reliable processing of application events across distributed systems
- Led the development of a **real-time event verification system** to validate incoming event data against defined requirements and **automate ticket generation for invalid or incomplete instrumentation**, improving reliability of downstream data workflows
- Designed and developed **backend services for an event instrumentation system** to monitor, validate, and manage event definitions and blueprints across Android, iOS, and Web applications

- Designed and developed **REST APIs to receive, process, and route events** from the SriMandir website, enabling downstream analytics, data processing, and real-time event-driven workflows
- Designed and implemented **reverse ETL pipelines** to **aggregate user attributes from GCP BigQuery** and **ingest enriched user data into CRM platforms** through **scheduled batch jobs**
- Enhanced website tracking and guest-user identification workflows, **improving analytics funnel completion from 10% to 77%** and enabling more complete attribution of user activity
- **Integrations & Product Engineering:** Integrated event and user-data flows with AppsFlyer, Mixpanel, MoEngage, Interakt, LeadSquared, and BigQuery to **power product analytics, user attribution, marketing automation, and CRM workflows**
- Built an end-to-end WhatsApp customer support rating workflow, integrating customer interactions with internal systems and **increasing user feedback rates from 38% to 79%**
- Built the **ClickToCall feature** and a bidirectional Exotel–LeadSquared call-log integration using AWS Lambda, **automating call and metadata synchronization and saving the sales team 4+ hours of manual work per day**
- **Infrastructure:** Optimized AWS ECS infrastructure by improving resource utilization, reducing cloud infrastructure costs by **$1,500 per month** without performance degradations
- Refactored the API Gateway authorizer Lambda into a structured and maintainable codebase, reducing development effort and improving the maintainability of authorization logic

Tech Stack: Go, PostgreSQL, MongoDB, Redis, AWS (Kinesis, Lambda, ECS, CloudWatch), GCP BigQuery, Bitbucket CI/CD

AppsForBharat January 2023 – April 2023
Software Engineering Intern *Bengaluru, India*

- Built an event property aggregator service in Go to provide centralized access to event properties across microservices, simplifying data access across the platform.
- Developed a reminder feature for festive task management, implementing backend workflows and data persistence using Go and MongoDB.

Tech Stack: Go, MongoDB

Projects

A/B Experiment | *GoLang (Gin), Postgres*

- Lets you create and run A/B experiments on your product
- Quickly create eligibility for your experiments, rollout strategy and start experimenting things

Weekly Planner | *Python (Django), Typescript, React, Postgres, Jev*

- A smart weekly planner which keeps track of your personal and professional goals
- Lets you plan your week into tasks based on your preferences and schedule for that particular week

Technical Skills

AI coding tools: Cursor, ClaudeCode, Codex
Languages: Go, Python, SQL, JavaScript, TypeScript
Technologies: Django, Gin, PydanticAI, React.js
Data and messaging: PostgreSQL, Redis, MongoDB, BigQuery, Data Pipelines (Streaming event pipelines and batch processing), ETL/Reverse ETL
Architectures: Microservices, event-driven
Cloud and infra: AWS - Kinesis, ECS, EC2, Lambda, CloudWatch; GCP - CloudRun, CloudSQL, MemoryStore, Cloud Storage, BigQuery